Economical cathode for lithium-ion batteries

Researchers led by Hailong Chen at Georgia Tech have developed a low-cost cathode material made from iron chloride (FeCl3), which could significantly improve lithium-ion batteries (LIBs). This breakthrough could lower battery costs by 98%, especially benefiting electric vehicles (EVs), where battery expenses make up about 50% of the total price. The FeCl3 cathode, using abundant materials like iron and chlorine, offers a sustainable and affordable alternative to expensive metals like cobalt and nickel. It also enhances energy storage when paired with solid electrolytes, making all-solid-state batteries safer and more efficient. Commercial viability is expected within five years.

Zhantao Liu with the new low-cost cathode that could revolutionise lithium-ion batteries and the EV industry (Photo by Jerry Grillo)

OLED-CMOS probes for cochlear implants

Researchers from the Fraunhofer IPMS and Max Planck MPI-NAT have advanced optical stimulation technology for future cochlear implants. This discovery uses optogenetics to control genetically modified cells via light, offering precise stimulation where electrical cochlear implants (eCIs) fall short. OLED-on-silicon technology enables more targeted stimulation, improving sound quality and frequency resolution. While challenges like flexibility and biocompatibility remain, the technology shows promise for cochlear implants and potential applications in other medical devices like pacemakers and pain management systems.

Chiplet with wireless terahertz technology

Researchers from Australia and the US are investigating new ways to improve chip-scale communication by using terahertz (THz) frequencies. This approach enables chiplets to communicate wirelessly through small antennas. The team applied Floquet engineering, a technique from quantum physics that manipulates electron behaviour in response to high-frequency signals, enhancing the system's responsiveness and ability to discern THz signals amid noise. Their experiments utilised a two-dimensional semiconductor quantum well (2DSQW). Additionally, the researchers introduced a dual-signalling architecture with two receivers that dynamically adjust the system's reference voltage based on detected noise levels, lowering error rates compared to single-receiver systems and enhancing signal decoding accuracy.

Helping robots focus on key objects

MIT engineers have developed Clio, a system that enables robots to make human-like, task-relevant decisions. Named after the Greek muse of history, Clio processes natural language task descriptions and interprets its environment by remembering only the most relevant details. Using deep-learning and 'open-set' recognition, it analyses millions of image-text pairs to understand scene elements. Applying the 'information bottleneck' principle, Clio segments and processes key information for specific tasks. Tested with Boston Dynamics' Spot robot, Clio successfully organised spaces and identified target objects. Future improvements aim to handle more complex tasks and incorporate advanced visual scene representations.

From left to right: team members Lukas Schmid, Nathan Hughes, Dominic Maggio, Yun Chang, and Luca Carlone (Credit: Andy Ryan)

TeSeO compound for semiconductor functionality

Researchers at City University of Hong Kong have advanced inorganic semiconductors by improving the mobility of positively charged carriers, or 'holes,' through an innovative blending strategy. They developed a new compound, tellurium-selenium-oxygen (TeSeO), outperforming conventional P-type semiconductors. TeSeO thin-film transistors and flexible photodetectors exhibit high stability and mobility, overcoming long-standing challenges in P-type materials. By engineering its band structure, the team achieved tuneable bandgaps ranging from 0.7eV to 2.2eV, enabling applications in ultraviolet, visible, and short-wave infrared technologies. This shows high-performance, cost-effective semiconductor devices and opens doors to future innovations.